Fiscal Year:	FY 2010	**Task Last Updated:**	FY 01/29/2010
PI Name:	Yokota, Hiroki Ph.D.		
Project Title:	Integrated Stress Response in Space		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	46202	**Congressional District:**	7
Comments:	NOTE: PI moved back to Indiana University from Rensselaer Polytechnic Institute in January 2011 [Ed., 01/26/2011]. PI previously at Indiana University until August 2010 [Ed. 09/2010]		
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Space Biology-FSB-Animal Physiology: NNH09ZTT003N
Start Date:	04/01/2010	**End Date:**	09/13/2010
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Sun, Sidney C	**Contact Phone:**	650-604-4835
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/13/2010, per PI's move to Rensselaer Polytechnic Institute (Ed. per A. Chu/ARC)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX10AE37G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A central objective of the proposed project is to study about the effects of integrated stress response (ISR) in space using bone growth and loss as a physiological target. It is known that various environmental stresses including radiation, oxidative stress, nutrient limitation, and stress to the endoplasmic reticulum (ER) induces ISR. ISR leads to a pro-survival pathway (rescue program) for reducing cellular injury or, conversely, a cell death pathway for removal of damaged cells. Unloading in space is a unique form of environmental stress to bone and surrounding tissues, but its physiological responses in connection to other stressors is poorly understood. Through a better understanding of linkage between ISR and unloading in space, we believe that development of a novel therapeutic strategy is possible for preventing bone loss in space. In this project, we address a set of questions: Does unloading evoke reprogramming of protein synthesis through ISR-linked modification of a particular protein named eukaryotic initiation factor 2a (eIF2a)? If so, what eIF2a driven events will control a choice of an ISR-linked pro-survival or cell-death pathway? We hypothesize: (1) Transient unloading induces an ISR-linked pro-survival program by activation of eIF2a modification and reduction in protein synthesis. (2) Continuous unloading induces oxidative stress and cell death through accumulation of stress to the ER, but this continuous stress effect can be reduced through sustained modification of eIF2a with administration of the chemical named salubrinal. To test the above hypotheses and examine a potential role of ISR-driven signaling under unloading, we propose three specific aims using a mouse hindlimb suspension model: Aim 1: Evaluate modification of eIF2a and its modifiers (for instance, PERK and GCN2 genes) as well as efficiency of protein synthesis in response to transient unloading Aim 2: Evaluate gene regulatory mechanisms linked to oxidative stress and stress to the ER in response to continuous unloading Aim 3: Examine the effects of administration of salubrinal (pharmacological agent that maintains modification of eIF2a) on the ISR-linked pro-survival pathway and bone metabolism under continuous unloading The project will employ various molecular and cellular techniques including real-time polymerase chain reactions, Western blot analysis, enzymatic activity assays as well as bone histology. To test specific signaling, we will use pharmacological stimulators and inhibitors, and eIF2a-linked knockout mouse models. The proposed project is responsive to NASA's research emphasis on clarification of systemic and targeted molecular physiological effects of acute and/or chronic exposure to gravitational changes. We expect that the project will clarify linkage of ISR to unloading driven bone metabolism, and provide a potential therapeutic usage of salubrinal as a pro-bone agent.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project in FY2010. End date changed to 9/13/2010 per PI's move to Rensselaer Polytechnic Institute in August 2010 ; for subsequent reporting on this task, see task entitled, "Integrated Stress Response in Space--NNX10AU51G." [Ed. 09/16/2010] SUMMARY of RESEARCH PROGRESS for FY2010 (provided January 2011): Unloading is a cause of disuse osteoporosis to bedridden patients as well as astronauts. The ongoing study demonstrates that unloading elevated the phosphorylated level of eIF2a, while mechanical loading downregulated it. Since eIF2a phosphorylation is induced by various stress sources that induce an integrated stress response (ISR), the result suggests that mechanical loading potentially suppresses ISR-linked stress.
Bibliography Type:	Description: (Last Updated: 04/09/2013)